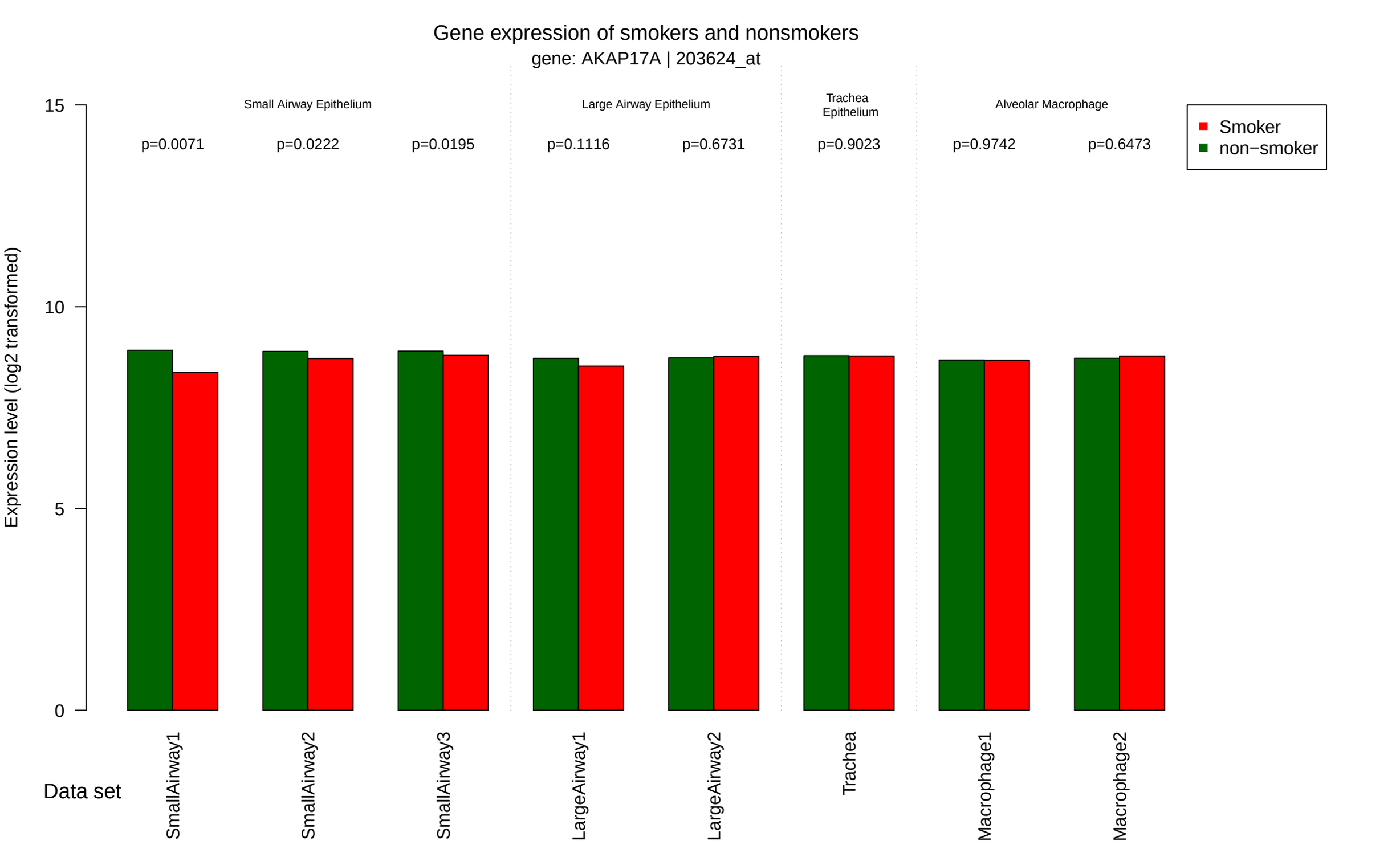
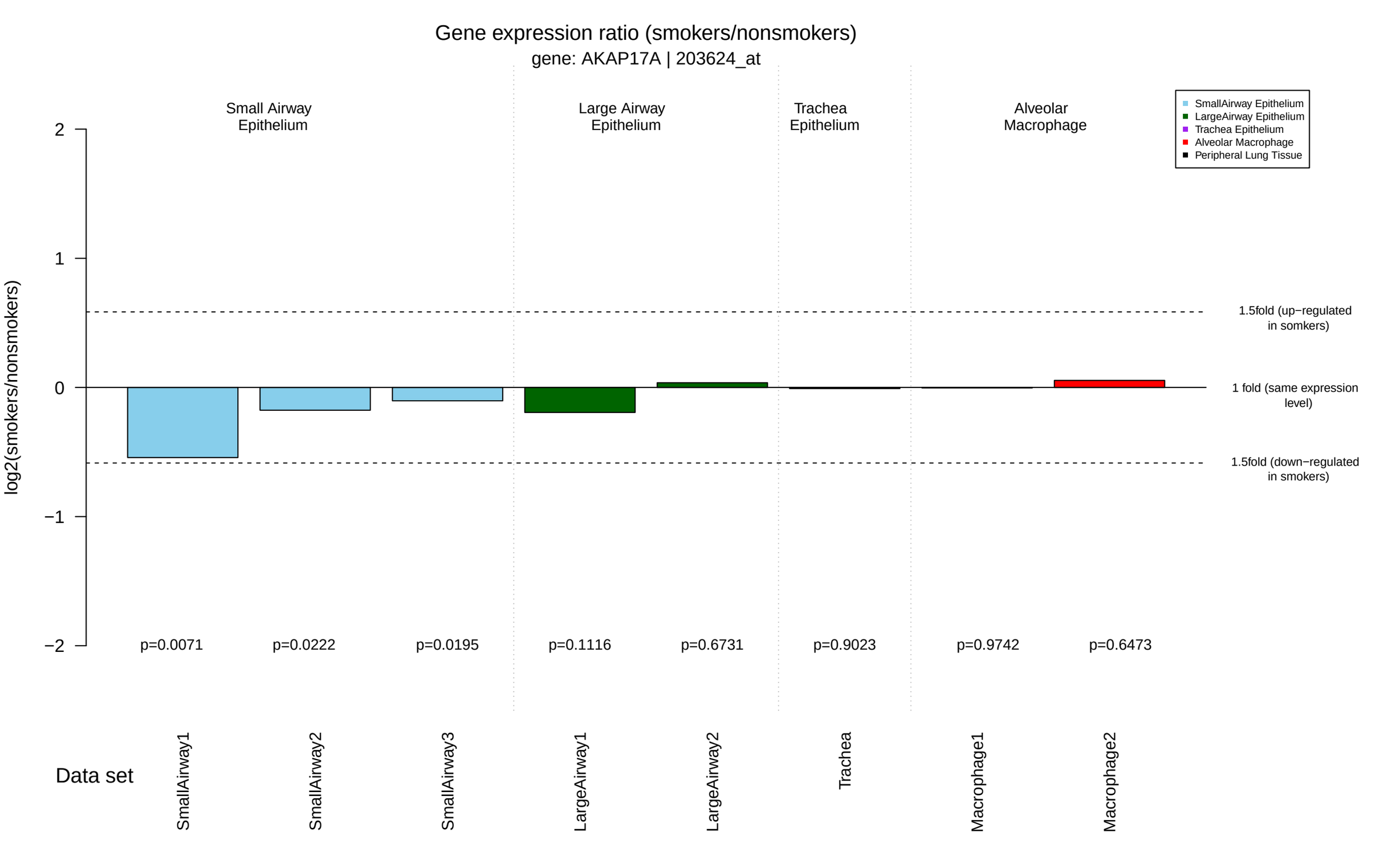
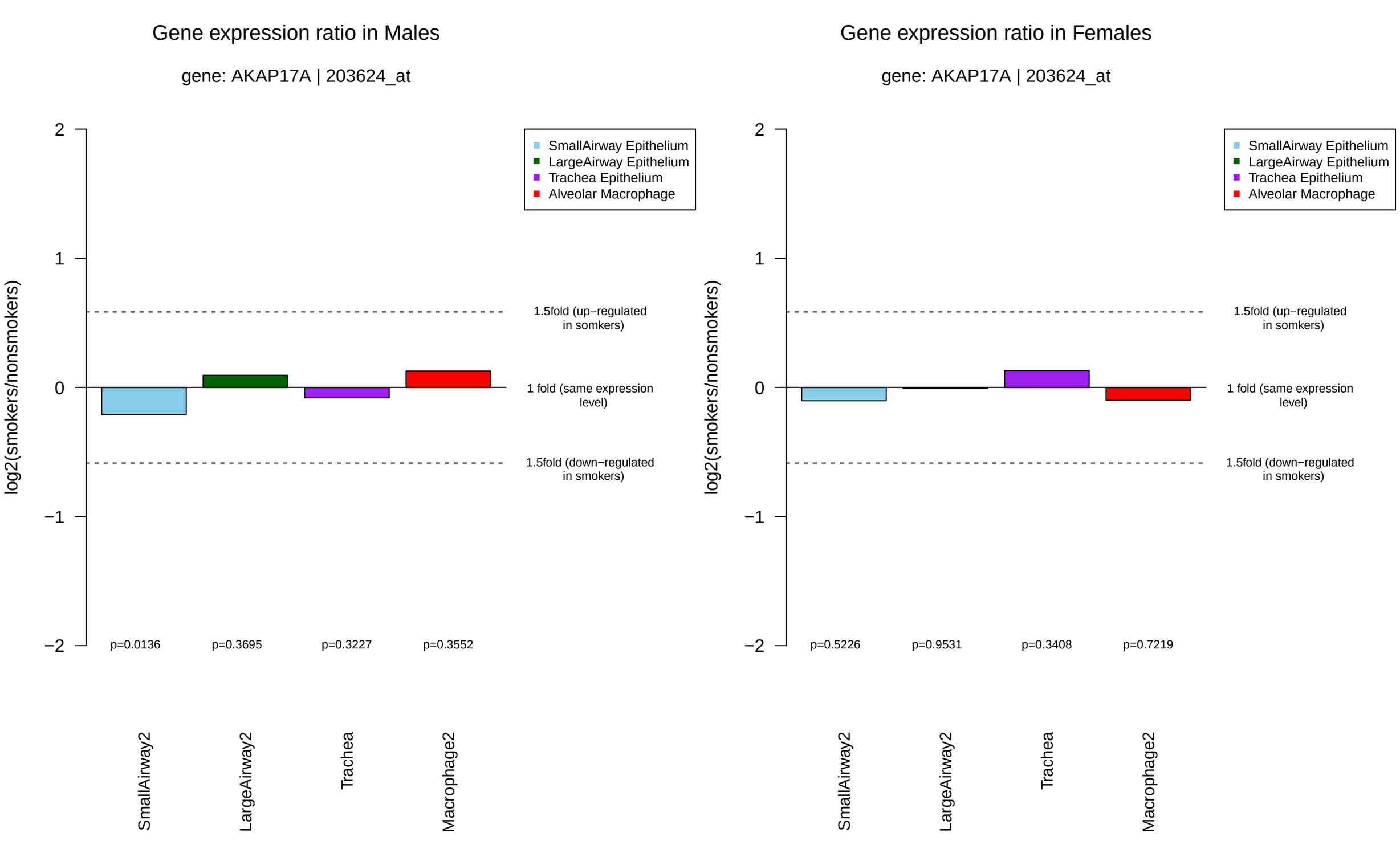
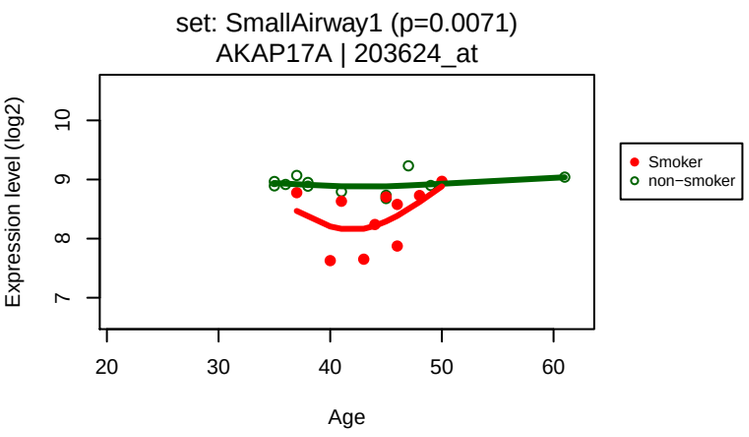
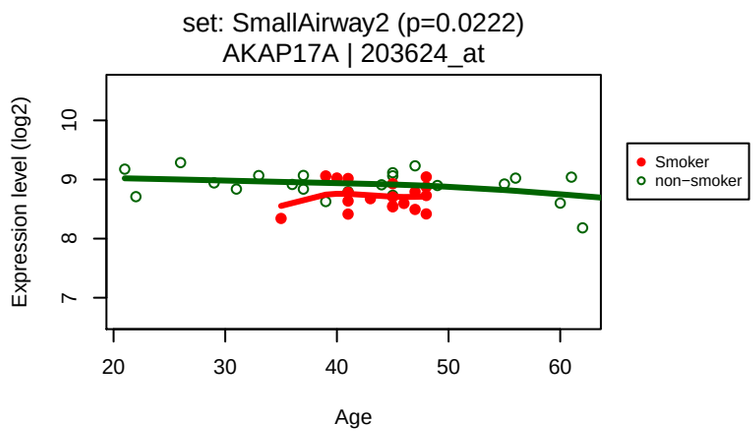
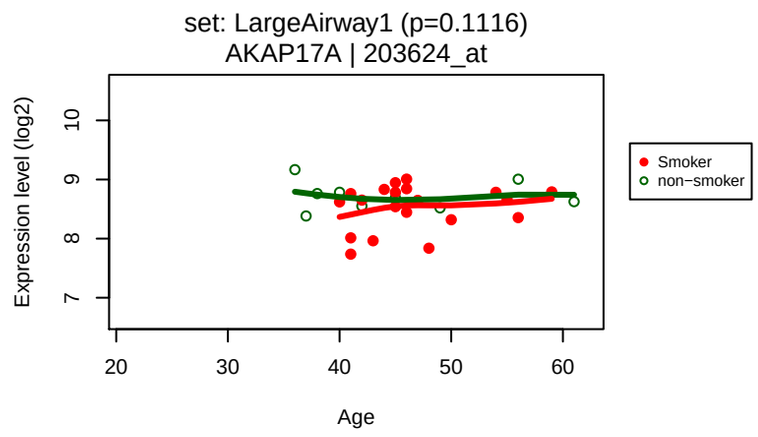

Gene expression of smokers and nonsmokers
gene: AKAP17A | 203624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0071
p=0.0222
p=0.0195
p=0.1116
p=0.6731
p=0.9023
p=0.9742
p=0.6473
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

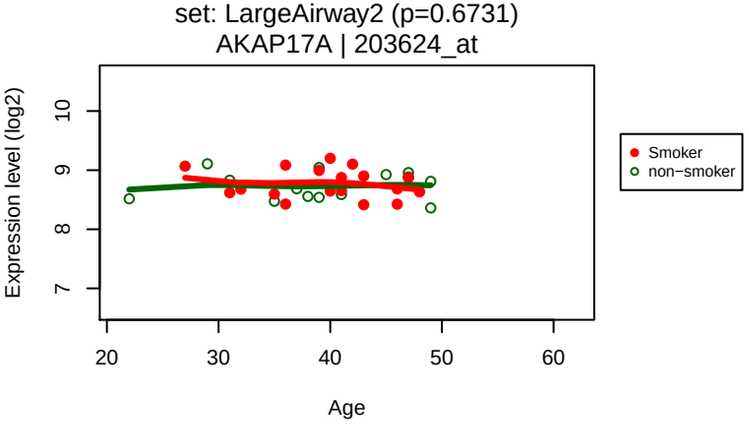

Gene expression ratio (smokers/nonsmokers)
gene: AKAP17A | 203624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0071
p=0.0222
p=0.0195
p=0.1116
p=0.6731
p=0.9023
p=0.9742
p=0.6473
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

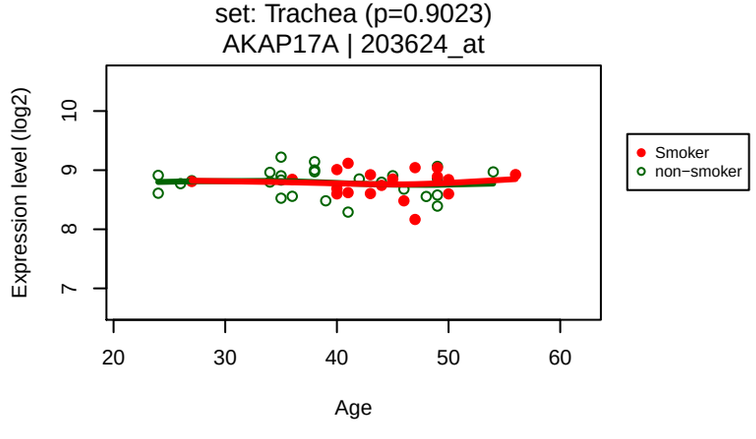

Gene expression ratio in Males
gene: AKAP17A | 203624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0136
p=0.3695
p=0.3227
p=0.3552
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP17A | 203624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5226
p=0.9531
p=0.3408
p=0.7219
SmallAirway2
LargeAirway2
Trachea
Macrophage2

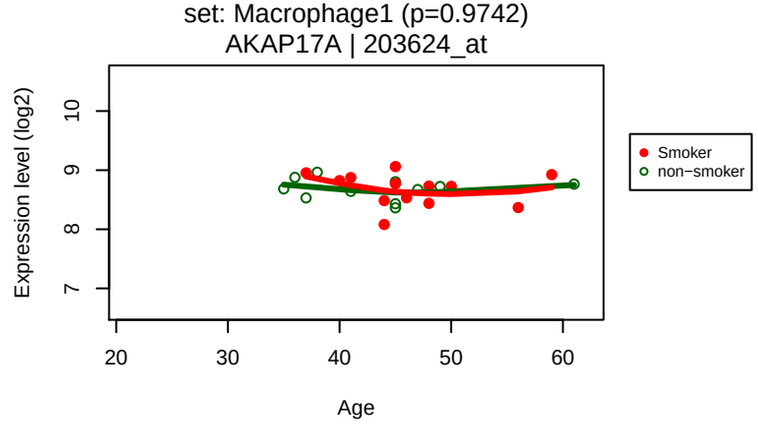

set: SmallAirway1 (p=0.0071)
AKAP17A | 203624_at
Expression level (log2)
Age
Smoker
non−smoker

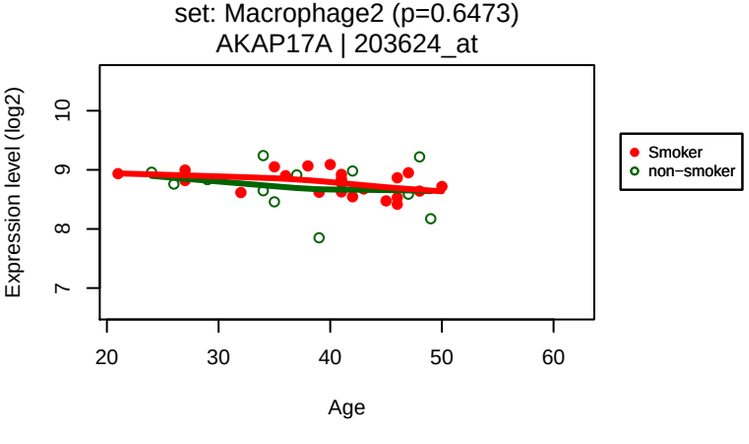

set: SmallAirway2 (p=0.0222)
AKAP17A | 203624_at
Expression level (log2)
Age
Smoker
non−smoker

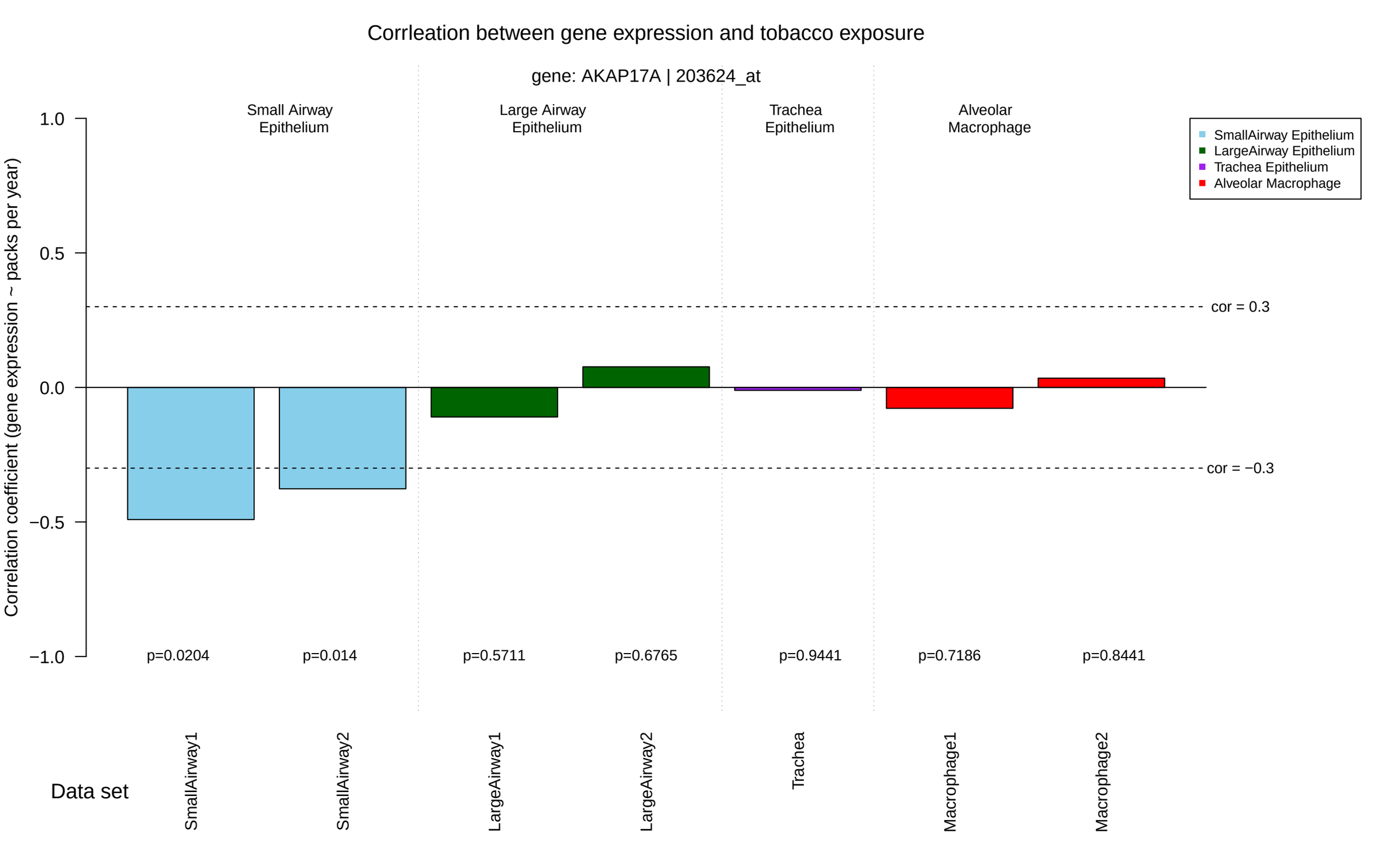

set: LargeAirway1 (p=0.1116)
AKAP17A | 203624_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6731)
AKAP17A | 203624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9023)
AKAP17A | 203624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9742)
AKAP17A | 203624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6473)
AKAP17A | 203624_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP17A | 203624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0204
p=0.014
p=0.5711
p=0.6765
p=0.9441
p=0.7186
p=0.8441
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2